

Day 20.5, Unit 2 October 13

Agenda

1. Pick up new Agenda, Turn in old Agendas (put today's agenda on new Agenda Sheet)
 2. 2.5 Demographic Transition Epidemiological Model Notes
- Reminder: Finish Population Pyramid Analysis and turn in next class!
 - Today's Objective: Describe Demographic Transition and Epidemiological Models

Video Warm Up

1. Describe the conditions of London in each stage of the DTM (1 - 4).
2. Why is the population still increasing if birth rates have dropped?
3. Why is it beneficial for developed countries to encourage advancements in developing countries?



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What is the
demographic
transition model?



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“The Demographic Transition Model explains the transition from high birth and death rates to low birth and death rates as a country develops from a pre-industrial to an post-industrial economic system.



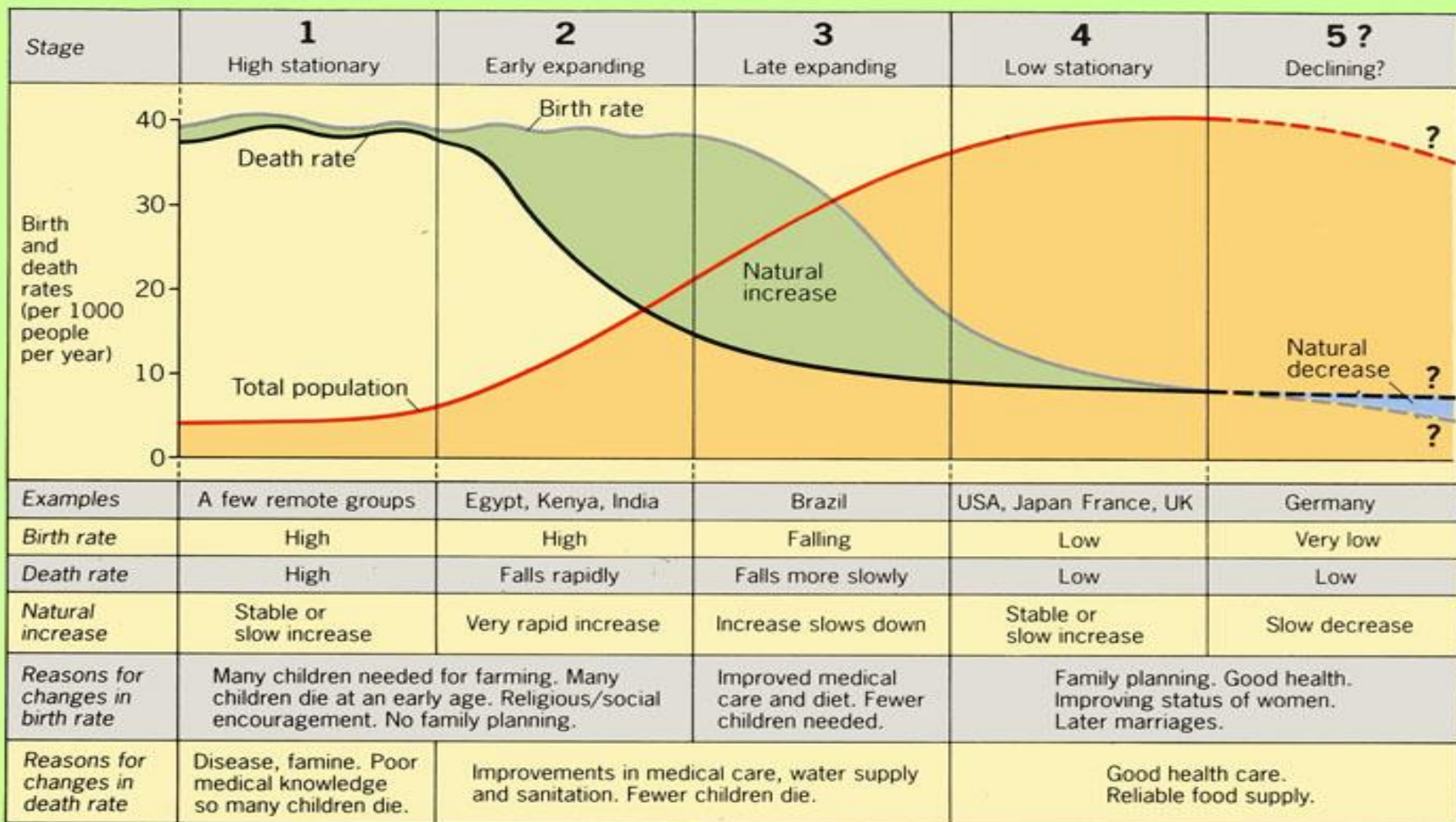
Stage 1:
Low
Population
Growth

Stage 2:
High
Population
Growth

Stage 3:
Moderate
Population
Growth

Stage 4:
Low
Population
Growth

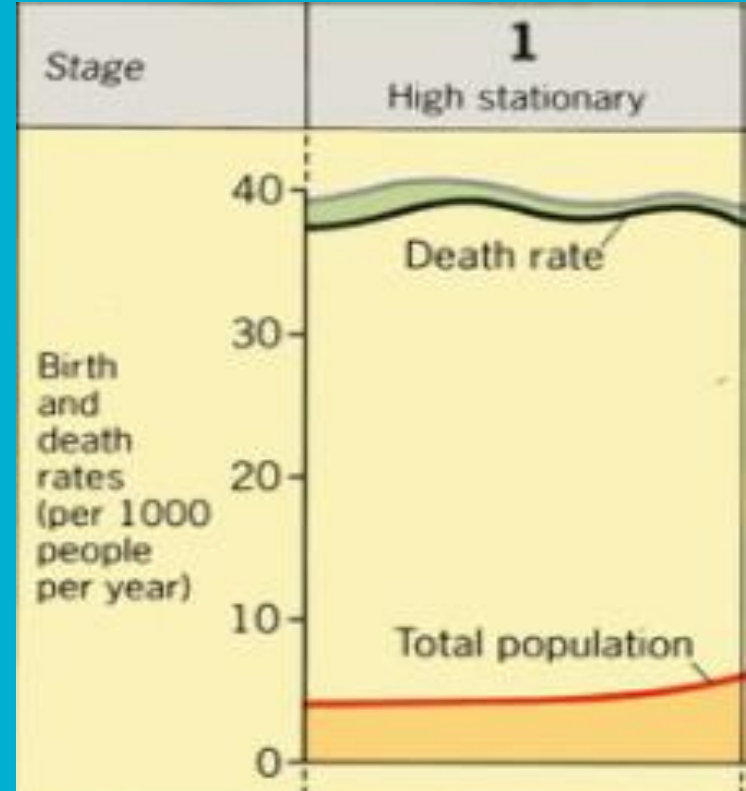
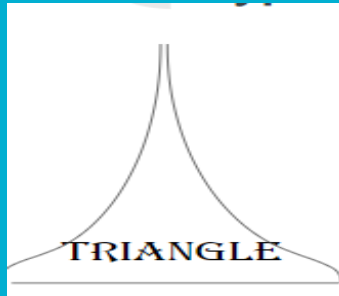
Stage 5:
Declining
Population
Growth



Stage 1: High Stationary Pop. Growth

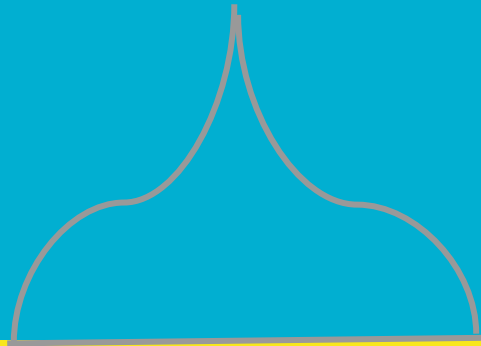
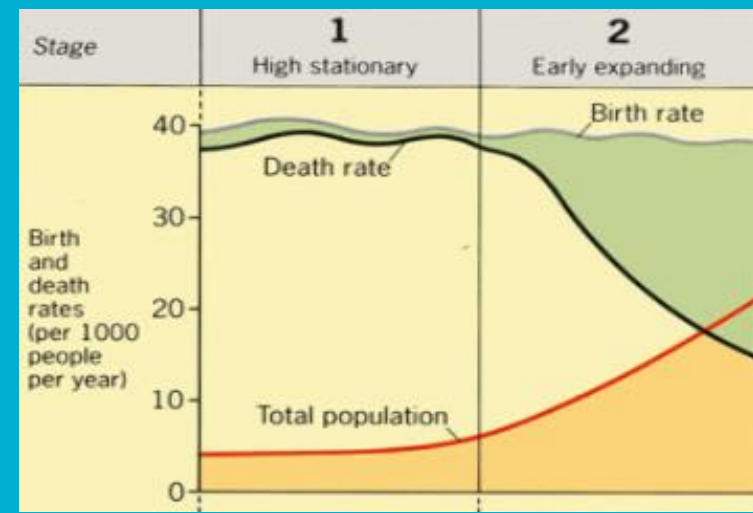
- *Pre-Industrial Revolution*
- Agrarian societies or tribal
- High stationary → High BR = High DR
- Most Modern countries are beyond this stage due to diffusion of modern medicine

Narrow Triangle/High
Stationary Growth



Stage 2: Early Expanding

- *Industrial Revolution* Begins
- Healthcare and technology = longer life
- Death rate drops but birth rate stays high
- Children are still needed/wanted to work in the fields or for the family
- Little (if any) family planning / pregnancy prevention = High Pop growth
- Ex. Kenya and Niger

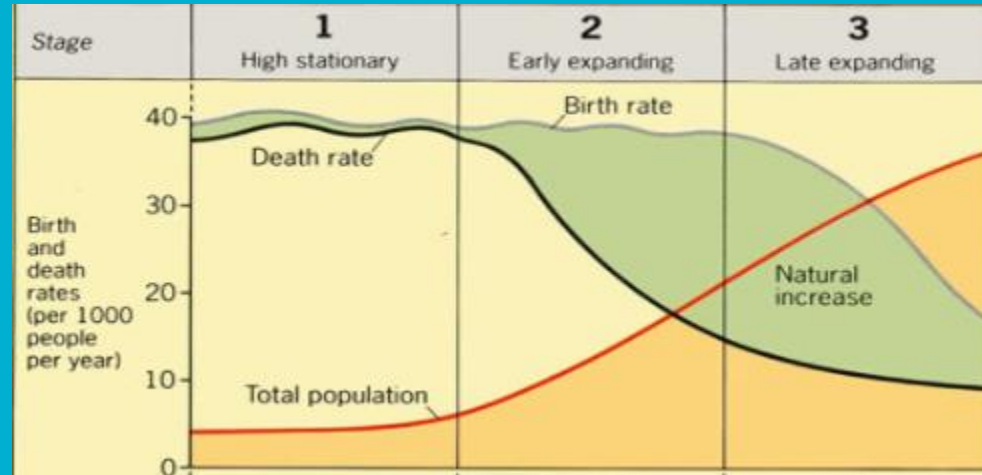
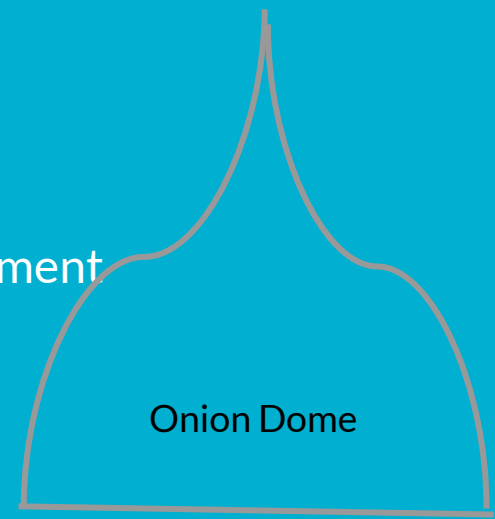


Not a full Onion Dome but the skinny triangle is starting to bulk up a little in the middle b/c death rates are falling- looks at little like a Hershey's Kiss

High Pop Growth!!!

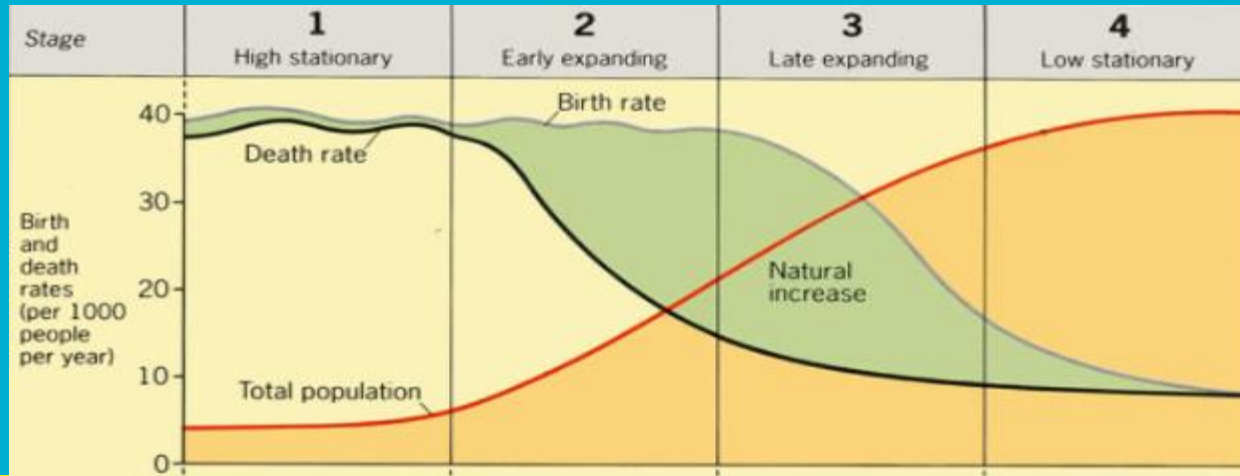
Stage 3: Late Expanding

- *Economic Development* - further into industrial development
- CBR declining and CDR stays low
- High rate of population growth
- Starting to have Smaller families
- Better quality of life + children in school
- This is the “transition” of the DTM
- Ex. Brazil

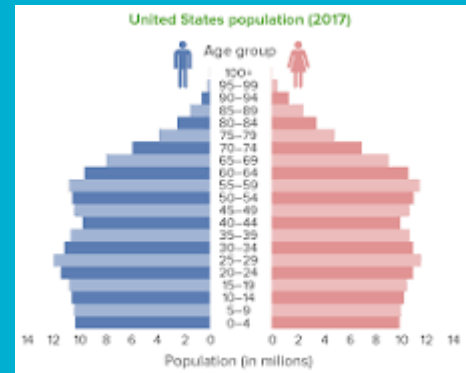


Stage 4: Low Stationary Growth

- Post Industrial Countries (MDCs/Core)
- Low BR = Low DR - population may fluctuate depending on economy
- Population is stable or zero population growth (~2.1 TFR)
- Ex. USA

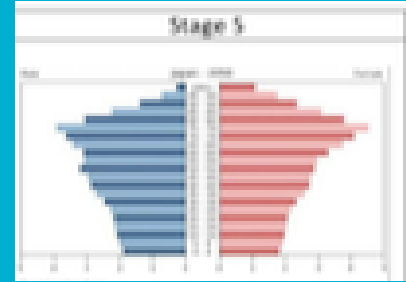


The Box



Stage 5: Declining

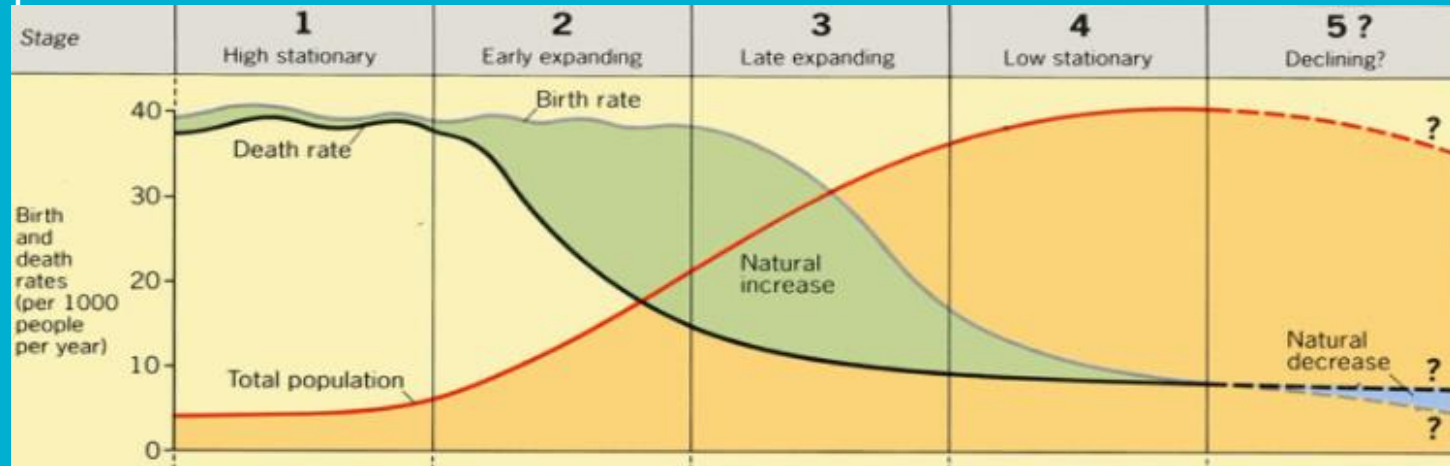
- Post-Industrial Countries (MDC/Core)
- CBR drops below C DR = negative population growth (decline)
- Rate of natural increase (RNI) is less than 0 (RN Decrease)
- TFR below replacement level



The Cup

Equation for Rate of Natural Increase (RNI)

$$(\text{BirthRate}/1,000) - (\text{DeathRate}/1000) = (\text{BR}-\text{DR})/1000 = \text{RNI} \%$$



*Rate of Natural Increase

- measure of how quickly a population is growing or declining.
 - Calculated using birth and death rates
 - does not factor in population change resulting from immigration or emigration

Equation for Rate of Natural Increase (RNI)

$(\text{BirthRate}/1,000) - (\text{DeathRate}/1,000) = (\text{BR}-\text{DR})/1,000 = \text{RNI} \%$

Examples of Calculating RNI

Let's look at a couple of examples. In [Nicaragua](#), the birth rate is 20/1,000 and the death rate is 5/1,000. Following the formula for finding RNI...

$20/1,000 (\text{BR}) - 5/1,000 (\text{DR}) = 15/1,000$

15 divided by 1,000 = 0.015 which when converted to a percentage equals 1.5% (Percentage Conversion: $.015 \times 100$)

So Nicaragua's rate of natural increase is 1.5%.

An RNI percentage larger than zero indicates a growing population. In some countries, like Germany, the death rate is larger than the birth rate, which causes a negative percentage. In the case of Germany, the birth rate is 9/1,000 and death rate is 11/1,000, resulting in an RNI of -0.2%. A negative RNI indicates a decreasing population size.

Time to play a game

Instructions:

- 1) Play the game as many times as it takes to get all items correct.

[DMT Game](#)

Core, Semi-Periphery, and Periphery



DTM & Semi Periphery - Periphery

- The periphery & semi-periphery countries seem to be stalled in the transitional phases (between stage 2/early expanding & stage 3/late expanding)
- “Demographic trap” - fertility remains high while mortality rate drops - still see large families even though the country is developing and health care/education is increasing

DTM Limitations

- Assumes **wealth** is distributed equally
- Does not take into account **immigration**
- Ignores culture/cultural **barriers impacting birth/death rates**
- Assumes all countries will go through the same process

In reality...

- DTM is best used when analyzing economic growth of **CORE** countries
- **Industrialization** is key and peripheral countries do not follow traditional industrial growth/patterns
- *Remind you of a theory of development? Which one: World Systems Theory OR Rostow's Modernization Model*

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What is the
Epidemiologic
Transition Model?



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“The **Epidemiologic Transition Model** focuses on **distinctive causes of death** in each stage of the demographic transition.”



Stage 1

- The stage of **pestilence and famine**
- Example: The Black Death
- Very Low Life Expectancy

Stage 2

- The stage of receding **pandemics**
- Example: Cholera
- Relatively low life expectancy

Stage 3

- The stage of **degenerative** and **man-made diseases**
- Examples: Alzheimer's, Cancer and Cardiovascular Disease
- Longer Life Expectancy

Stage 4

- The stage of **delayed degenerative diseases**
- Example: Cancer & disease is slowed or stopped by medical advances.
- Very Long Life Expectancy

Stage 5 (Possible)

- **Reemergence of infectious and parasitic diseases.**
- Reasons: Evolution of pathogens/viruses, Poverty, Improved Travel increases disease contagion; could impact Life



Billy was a gravedigger...

Choose which DTM stage being described in each scenario